

ROBLIN, MB, Canada

WMO: 715530

Lat: 51.1833N

Lon: 101.3628W

Elev: 1770

StdP: 13.78

Time Zone: -6.00 (NAC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.3	-23.8	-34.5	0.8	-28.2	-29.5	1.1	-23.7	27.9	13.4	24.8	12.4	5.9	210	0.719

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.5	82.9	67.8	79.8	66.0	77.0	64.4	71.3	79.5	68.9	76.6	66.7	73.9	11.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.7	112.7	75.9	66.2	102.9	73.1	63.9	95.0	71.2	36.4	79.8	34.2	76.5	32.3	74.0	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.6	21.7	19.5	-35.8	89.5	4.7	3.1	-39.2	91.7	-42.0	93.5	-44.6	95.2	-48.1	97.4
			-35.1	75.4	5.0	2.6	-38.7	77.3	-41.7	78.8	-44.5	80.3	-48.2	82.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	35.0	2.6	5.5	18.4	36.9	49.9	59.1	63.9	61.8	53.0	38.7	21.3	6.8
	DBStd	24.66	15.14	13.95	13.71	11.23	8.61	6.32	5.49	6.10	8.00	9.07	12.23	14.48
	HDD50	6840	1470	1245	981	406	110	7	0	1	55	364	861	1340
	HDD65	11070	1935	1665	1446	842	470	194	86	135	366	815	1311	1805
	CDD50	1362	0	0	0	14	108	279	432	367	147	15	0	0
	CDD65	116	0	0	0	0	3	17	53	36	7	0	0	0
	CDH74	1340	0	0	0	3	80	186	524	423	120	4	0	0
	CDH80	255	0	0	0	0	15	27	95	98	20	0	0	0
	WSAvg	9.3	9.1	8.7	9.8	10.4	11.0	9.6	8.1	8.1	9.2	9.9	9.5	8.7
Precipitation	PrecAvg	19.9	0.9	0.7	0.9	1.1	2.2	3.5	3.0	2.9	1.7	1.4	0.8	0.9
	PrecMax	27.2	1.7	1.5	2.5	3.1	5.0	6.5	6.5	5.2	2.8	3.9	2.0	2.0
	PrecMin	13.0	0.2	0.0	0.2	0.1	0.6	0.9	0.9	0.8	0.2	0.4	0.1	0.5
	PrecStd	3.3	0.4	0.4	0.6	0.7	1.3	1.7	1.3	1.3	0.7	0.7	0.6	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.1	37.3	49.9	71.8	81.6	83.3	87.1	88.7	82.7	71.8	51.5	38.2
		MCWB	33.7	34.4	42.1	54.6	59.0	68.2	72.0	68.7	64.4	56.7	44.7	35.0
	2%	DB	31.5	33.4	43.3	64.5	75.4	79.1	82.4	82.6	77.7	63.2	44.8	33.6
		MCWB	29.9	32.0	39.0	49.9	56.0	64.4	70.2	67.3	62.3	51.0	39.7	31.5
	5%	DB	28.0	29.7	39.1	59.5	70.8	75.4	79.4	78.7	72.3	57.2	40.0	29.5
		MCWB	26.8	28.5	35.9	47.0	54.6	62.0	68.0	65.5	59.7	48.5	36.6	27.9
	10%	DB	24.1	25.7	35.8	54.2	66.1	72.0	76.6	75.2	67.4	52.2	36.4	26.0
		MCWB	23.2	24.8	33.7	44.7	51.8	60.4	66.5	64.0	57.1	45.7	34.1	24.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.9	35.0	44.1	55.3	62.5	72.1	76.2	73.2	68.8	57.1	45.5	35.2
		MCDB	35.8	36.6	47.8	68.2	72.8	80.2	83.3	81.1	77.0	69.2	51.1	37.4
	2%	WB	30.1	32.0	39.0	51.4	58.9	67.5	72.2	70.1	64.7	52.9	40.4	31.4
		MCDB	31.6	33.4	42.7	61.2	70.0	74.0	79.2	79.1	73.9	61.2	43.8	33.6
	5%	WB	26.7	28.4	36.2	48.2	56.3	64.7	69.9	67.6	61.4	49.5	37.0	27.8
		MCDB	27.7	29.4	38.9	57.7	67.9	72.1	77.9	75.7	70.0	55.7	39.8	29.2
	10%	WB	22.8	24.5	33.8	45.2	53.8	62.3	67.7	65.6	58.0	46.6	34.2	24.6
		MCDB	23.5	25.5	35.9	53.6	63.8	69.7	74.8	73.1	66.6	52.5	36.4	25.5
Mean Daily Temperature Range	5% DB	MDBR	16.4	17.6	17.5	19.2	21.7	19.4	20.5	21.8	20.8	17.0	14.1	15.1
		MCDBR	16.7	16.2	17.1	27.0	29.0	24.8	24.9	27.3	28.7	26.7	18.3	16.8
		MCWBR	16.2	14.8	13.4	15.0	13.9	12.4	14.0	13.9	15.0	16.1	13.8	15.5
	5% WB	MCDBR	17.2	15.6	16.6	24.8	24.7	21.1	23.0	23.3	25.7	24.7	17.4	16.9
		MCWBR	16.3	14.2	13.1	14.6	13.3	12.4	14.2	13.7	14.7	15.4	13.4	15.4
Clear-Sky Solar Irradiance	taub		0.246	0.238	0.310	0.323	0.334	0.358	0.369	0.372	0.340	0.280	0.252	0.229
	taud		2.283	2.342	2.239	2.413	2.444	2.398	2.389	2.364	2.442	2.578	2.478	2.358
	Ebn at Noon		251	291	281	292	292	285	279	272	270	273	253	244
	Edh at Noon		20	25	35	33	34	36	36	35	29	20	17	16
All-Sky Solar Radiation	RadAvg		403	773	1232	1608	1797	1836	1958	1595	1172	677	360	286
	RadStd		29	54	81	115	156	138	93	134	104	91	49	27

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	-1.10	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	-0.90	N/A	N/A	N/A	N/A

Nomenclature: See separate page